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DUNMAN HIGH SCHOOL

Preliminary Examination

Year 6

MATHEMATICS (Higher 1)

Paper 1

8864/01

17 September 2012

3 hours

Additional Materials:

Answer Paper

List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your Name, Index Number and Class on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer all the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, **attach the question paper to the front of your answer script.**

The total number of marks for this paper is 95.

For teachers' use:

Qn	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Total
Score														
Max Score	3	4	5	6	6	11	4	4	8	10	10	11	13	95

Section A: Pure Mathematics (35 marks)

- 1 Find algebraically, the set of values of k such that $kx^2 + kx + 2x + 4k > 0$ for all real values of x . [3]
- 2 A right-angled triangle ABC has a perimeter of 36 units. Given that AC is the hypotenuse of the triangle and AB is two units longer than BC , find the length of AC . [4]
- 3 Sketch the curve $y = \ln(2x - 3)$, stating clearly the equations of any asymptotes and the coordinates of any points of intersection with the axes. [2]

Express x in terms of y , and hence find the exact area of the region bounded by the curve $y = \ln(2x - 3)$, the x -axis and the line $x = 4$. [3]

- 4 (i) The gradient of a curve $y = f(x)$ at any point (x, y) is given by $\frac{2}{bx+3}$, where b is a constant. Given that $f'(2) = 2$ and the curve intersects the x -axis at $x = -1$, find the equation of the curve. [3]
- (ii) Hence solve $f(x+2) = f(x) + f(2)$. [3]
- 5 The curve C is given by $y = 28 + 20x - x^2 - 2x^3$.
- (i) Sketch C , indicating clearly the coordinates of any turning points and the points of intersection with the axes. [2]
- (ii) Hence state the range of values of k for which $2x^3 + x^2 - 20x = 28 - k$ has exactly one real solution. [1]
- (iii) Find the range of values of x such that $\ln(28 + 20x - x^2 - 2x^3)$ is defined. Hence find the range of values of x such that $\ln\left(28 + \frac{20}{e^x} - \frac{1}{e^{2x}} - \frac{2}{e^{3x}}\right)$ is defined. [3]

[Turn Over]

- 6 (a) A vessel is in the shape of an inverted right circular cone whose base radius is half the height of the cone. The vessel initially contains some water, which leaks out through a hole in the vertex at a rate of $3 \text{ cm}^3\text{s}^{-1}$. The volume of water in the vessel is $V \text{ cm}^3$ when the depth of water is $x \text{ cm}$.
- (i) Show that the volume of water in the vessel is given by $V = \frac{1}{12}\pi x^3$. [2]
- (ii) Hence find the exact rate of decrease of the depth of water when $x = 6$. [3]
- (b) A test rocket fires a small projectile into the air, which reaches a maximum height before falling back to the ground. The height of the projectile above ground, $h \text{ m}$, at $t \text{ s}$, is given by $h = -t^4 + \frac{28}{3}t^3 - 30t^2 + 36t + 1$.
- (i) Use differentiation to determine the value of t for which h is a maximum and state the corresponding value of h . [5]
- (ii) An observer notices the small projectile when t is between 1.5 and 3.5 inclusive. Find the exact greatest height attained by the projectile during this time interval. [1]

Section B: Statistics (60 marks)

- 7 X is a normal random variable with mean μ and standard deviation σ . A sample of 50 independent observations is taken. It is given that the probability that the sample mean is less than 128 is 0.15, and the probability that the sample sum exceeds 7500 is 0.25. Find the values of μ and σ , giving your answers to the nearest whole number. [4]
- 8 An office received a total of 810 loan applications on a particular day. The applications were categorised into business, housing and study loans. Due to a shortage of staff, the office was only able to process 30 applications on that day.
- (i) Describe how you would choose a systematic sample of size 30 from the loan applications. [2]
- (ii) Explain why a systematic sample may not ensure that all categories of loan applications were processed within that day. [1]
- (iii) Explain why a stratified sample may be preferred in the context of this question. [1]

[Turn Over]

- 9** The waiting time for a patient to see a doctor in a clinic is T minutes. The waiting times of a randomly chosen sample of 120 patients is summarised by $\sum(t-15) = 123$.

- (i) Find the unbiased estimate of the population mean. [1]
- (ii) What do you understand by the term “unbiased estimate”? [1]
- (iii) The standard deviation of T is known to be 4.47 minutes. The management of the clinic claims that the mean waiting time is 15 minutes. Carry out a test at the 5% significance level to determine whether the mean waiting time differs from 15 minutes. [3]
- (iv) In another sample, the waiting times of n patients (where $n \geq 50$) were recorded and it was found that $\bar{t} = 15.5$. Given that the clinic’s claim was rejected at the 6% significance level for a two-tailed test, find the least value of n . [3]

- 10** Data is gathered on how test scores, y , vary with the number of hours, x , that a student spent studying for a test. The data for the seven students is shown below.

Number of hours, x	1	2	3	6	12	24	36
Test score, y	20	33	49	52	70	80	95

- (i) Give a sketch of the scatter diagram for the data as shown on your calculator, and calculate the linear product-moment correlation coefficient. [2]
- (ii) Find the equation of the regression line of
 - (a) y on x , [1]
 - (b) x on y . [1]
- (iii) Explain what the gradient of the regression line of y on x represents. [1]
- (iv) Use a suitable regression line to estimate the number of hours spent on studying by a student who obtained a score of 60. Explain why it is inappropriate to use the other regression line. [2]
- (v) The data for a second sample of another 53 students is obtained. The regression lines of y on x and of x on y for the 53 students are given respectively as:

$$y = 1.6x + 35 \quad \text{and} \quad x = 0.5y - 14.$$

Calculate the mean score and mean number of hours spent on studying for the combined sample of 60 students. [3]

[Turn Over]

- 11** On a Friday evening, the class chairperson emails his 20 classmates to propose an outing to watch a newly released action thriller. The random variable X is the number of classmates who respond positively to his email that weekend.

It is assumed that X has the distribution $B(20, p)$ such that $0 < p \leq 0.5$.

- (i) Given that $P(X = 0 \text{ or } 19) = 0.1$, formulate an equation for p , and find its numerical value. [2]

Every weekday morning, the class chairperson draws a name from a bag containing the names of all the students in his class. The person whose name is drawn will be responsible for keeping the homeroom tidy. The class consists of 12 girls and 9 boys. The random variable Y is the number of days, out of a 5-day week, in which a girl is picked to be on duty.

- (ii) Explain in context clearly why the random variable Y is more suited to be modelled by a binomial distribution than X . [2]
- (iii) Find the probability that in a week, a girl is on duty more often than a boy. [3]
- (iv) Using a suitable approximation, find the probability that in a semester of 20 weeks, there are less than 11 weeks in which a girl is on duty more often than a boy in the week. [3]

- 12** (a) Two events A and B are such that $P(A) = 0.25$, $P(B) = x$, $P(A|B) = \frac{5x}{16}$ and $P(A \cup B) = \frac{7}{16}$, where $0 < x < 1$.

- (i) Find the value of x . [3]

- (ii) Determine whether A and B are independent. [1]

- (b) A man owns a wooden chest with three drawers. One drawer contains two gold coins, another contains one gold coin and one silver coin, and the last drawer contains two silver coins. A drawer is selected randomly and a coin is then selected randomly from that drawer.

- (i) Find the probability that a gold coin is selected. [1]

- (ii) Given that a gold coin is selected, find the probability that the chosen drawer is the one that contains two gold coins. [2]

The man's two sons take turns to select a coin each from a drawer in the same manner, starting with the older son.

- (iii) Given that the older son picked a gold coin from the drawer containing two gold coins, show that the probability that the younger son picked a gold coin is $\frac{1}{2}$. [1]

- (iv) Find the probability that both sons pick a gold coin each. [3]

[Turn Over]

- 13** Students seeking admission to Kingdom University and Island University are required to apply using their national examination scores. The scores of the prospective applicants for each university can be modelled by independent normal distributions with means and standard deviations as shown in the table.

	Mean Score	Standard Deviation
Kingdom University	500	100
Island University	480	60

- (i) Kingdom University only admits the top 30% of its prospective students. Find the minimum score a randomly chosen student must achieve in order to gain admission into this university. [2]
- (ii) Island University has admitted students who scored within one standard deviation from its mean. Find the percentage of prospective students Island University has admitted. [1]

A random sample of 5 prospective students of Kingdom University and a random sample of 5 prospective students of Island University are chosen.

- (iii) Find the probability that the difference in mean score between the prospective students of the two universities exceeds 50. [4]
- (iv) Find the probability that the total score of these 5 prospective students of Island University is more than 5 times the score of a randomly chosen prospective student of Kingdom University. [3]
- (v) Find the probability that each of these 10 prospective students achieves a score of at least 500. [2]

Explain why this probability would be less than the probability that these 10 prospective students achieve a total score of at least 5000. [1]

END OF PAPER